ABBA4 Implicit Single Projection

One symmetric projection around an unprojected triple jump

1 Unprojected fourth-order base

Let $\mathcal{A}_{q,t}$ be one signed endpoint-time ABBA map on two copies, with no intermediate diagonal projection. Define

$$\gamma = \frac{1}{2-\sqrt[3]{2}}, \qquad \delta = -\frac{\sqrt[3]{2}}{2-\sqrt[3]{2}}.$$

The complete base map is

$$\mathcal{C}_{h,t} = \mathcal{A}_{\gamma h, t+(\gamma+\delta)h} \circ \mathcal{A}_{\delta h, t+\gamma h} \circ \mathcal{A}_{\gamma h, t}.$$

Because $2\gamma + \delta = 1$ and $2\gamma^3 + \delta^3 = 0$, the palindromic composition cancels the leading third-order defect of a symmetric second-order base. The middle duration is negative, and the two copies pass continuously through all three factors.

2 One outer symmetric projection

For

$$E = \begin{pmatrix} I \\ I \end{pmatrix}, \quad P = \tfrac{1}{2}\begin{pmatrix} I & I \end{pmatrix}, \quad G = \begin{pmatrix} I & -I \end{pmatrix}, \quad N = G^T,$$

the reduced formulation is

$$M(\mu) = \mathcal{C}_{h,t}(Ez + N\mu), \tag{1}$$
$$r(\mu) = GM(\mu) + 2\mu = 0, \tag{2}$$
$$z^+ = P(M(\mu) + N\mu). \tag{3}$$

If J_j is the Jacobian of signed factor j, then $J_{\mathcal{C}} = J_3 J_2 J_1$ in execution order and

$$D_\mu r = G J_{\mathcal{C}} N + 2I.$$

Every residual evaluation traverses all three ABBA factors.

3 Configurations

The class exposes the same orthogonal axes as the other implicit ABBA methods: two projection formulations, Newton or Broyden, and physical, shared-time, or fully extended state. The selector values are global to the outer step. In the simultaneous formulation the corrected doubled output joins μ in the unknown; elimination gives the reduced residual above. State extension changes what is duplicated, but never introduces intermediate projection.

4 Difference from `ABBA4Implicit`

Both fourth-order classes use (γ, δ, γ), but `ABBA4Implicit` solves three projections and re-embeds a physical state between signed factors. This method solves one projection around the full unprojected composition. Since projection and factor evolution do not commute, the classes define different numerical maps rather than two costs for the same map.

5 Properties and limitations

The ideal method is symmetric and designed for order four. Geometric claims refer to the converged projection root; finite nonlinear tolerances perturb the map. Negative internal time increments require the potential and dynamics to be evaluable slightly outside the outer interval. Observer records preserve the three ordered factors so the accepted-map tangent can be reconstructed.

6 Implementation correspondence

The public class and outer residual are in `src/simulation/methods/abba/order4_implicit_single_projection.py`. The common ABBA stages and projection helpers live in the neighboring private modules. The companion architecture document under `../simulation/` gives dimensions, diagnostics, and runtime flow.